

THE FOOTHOLD LATTICE

A P R E P R I N T

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2 seeds × 5 lattice sizes, zero misses

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ABSTRACT

Can a continuous-state reservoir be *compiled* into a genuine finite-state automaton at negligible cost? The lab had already asked this once with the obvious tool — quantizing every coordinate of the state to b bits — and the answer was a clean no: determinism never appeared at any resolution from 1 to 23 bits, because a coarser grid still leaves astronomically many representable states. This study changes one thing: instead of rounding coordinates independently, the state is snapped, every step, to its nearest vector in a **fixed codebook of K states the pond has actually visited** — a hard cap on the cardinality of the whole state vector. That one change flips the answer. (1) At *every* tested $K \in \{16, 64, 256, 1024, 4096\}$, on both seeds — 10 of 10 cells — the pond freezes into a real automaton: identical recent-character suffixes of depth 7–11 produce the identical state at every one of 40,000 probed positions. (2) The dial is smooth: prediction cost degrades monotonically as the codebook shrinks, 8/8 adjacent comparisons, no cliff, no reversal. (3) The largest tested lattice is nearly free: $K=4096$ costs only 0.053–0.057 bits over the unquantized float32 reference — the cheapest tested point on the dial is also the one that compiles.

Keywords reservoir computing · vector quantization · finite-state automata · state-space discretization · determinism · pre-registration

§1 The question

A reservoir's state is a point in a 1024-dimensional continuum; its dynamics are checkable only statistically. A finite-state machine, by contrast, can be tabulated, audited, and proven things about. The lab's earlier crystal-pond study (run ledger #7) posed the compilation question with per-coordinate rounding and got a striking negative: bits-per-character cost melted smoothly below $b \approx 6$ bits and was *exact* to float32 from $b=8$ — but short-memory determinism stayed at exactly 0.0 at every b from 1 to 23. Rounding re-injects a small perturbation each step, and a b -bit grid over 1024 coordinates still has vastly more representable states than any run visits; the pond never repeated itself in 40,000 sampled states. The goat-family study of descent trails (Goat Descent) had meanwhile left an

unbuilt image in the lab's notes: *a pond whose state walks a fixed lattice of footholds*. This study builds that pond, asking: is a hard cardinality cap — not a finer or coarser grid — what forces a reservoir to become a machine?

§2 The machine

The base pond is the house champion tanh reservoir at its best measured operating point ($N=1024$, $\rho=0.8$, sparse fanin-10 mixing, house input map, leak 1.0) [3] — the same anchor cell the crystal-pond study quantized. One line is added to its update:

```
# C = codebook: K states the unquantized pond visited
#   during a fixed 100k-char warmup, sampled once,
#   frozen for the whole run (no k-means, no synthesis)
x ← argmin_{c ∈ C} ||x - c|| # after every update step
```

The snapped state feeds the next step, so the closed-loop dynamics live entirely on the K footholds — nearest-neighbor vector quantization [1] of the state, with the codebook drawn from the pond's own visited set rather than fit to it. K sweeps {16, 64, 256, 1024, 4096}. Instruments are copied verbatim from the crystal-pond study for direct comparability: ridge and logistic validation bpc on standard text8 splits [4] (2M train characters), and the determinism probe (T5 in the lab's internal instrument numbering): over 40,000 sampled positions, group states by their length- k character suffix and ask whether every multi-sample suffix class lands in exactly one state. k^* is the smallest depth at which that fraction reaches 1.0, judged only where at least 20 multi-sample classes support it — at $K=4096$ (seed 0), $k^*=10$ rests on 2,172 such classes, not a thin slice. A contract cell ($K=\infty$, no snapping) reproduced the plain unquantized pond bit-for-bit through the same code path in both pre-registration smokes.

§3 What we registered in advance

- **M1** (existence, per K , judged independently): does k^* come out finite at each of the five K 's? Named alternative, per K : the snap-to-nearest correction re-injects noise the way rounding did, and a small *cardinality* cap does not force determinism by itself. Any K freezing is newsworthy against the 0/23 contrast.
- **M2** (the dial): logistic val bpc strictly improves along the whole ladder $K = 16 \rightarrow 4096$, all four adjacent pairs (tie band ± 0.005). Named alternative: non-monotone — disclosed honestly, the pre-registration smoke had hinted k^* might dip at $K=256$, so cost might too.
- **MQ1** (the price): logistic val bpc at $K=4096$ lands in [2.6519, 2.80] — within 0.15 bits of the crystal-pond float32 reference (2.6519; quoted, not recomputed, and structurally not this dial's $K \rightarrow \infty$ limit, since a codebook pond only revisits stored anchors).

The registration also pre-stated the propagation rule: M1 firing at any K is provisional-only on a first seed; M1 *plus* MQ1 landing at the same K — a checkable automaton at negligible cost, the crystal-pond study's own named target — is EUREKA-tier under the lab's constitution, and triggers a replication

queued ahead of everything else before any entry in the findings book. That is what happened. The seed-1 rerun registered the same three claims (N1/N2/NQ1), each with its own unbundled promotion rule.

§4 What happened

Table 1. Both registered passes (run ledger #16 seed 0, #17 seed 1; values seed 0 / seed 1). "Used" = distinct codewords active in the closed loop. Float32 reference: logistic 2.6519, ridge 3.1073 (crystal-pond, ledger #7). Zero AUTO-VOIDs, 10/10 cells.

K	logistic val bpc	ridge val bpc	k*	used
16	3.7762 / 3.7695	4.0528 / 4.0668	8 / 7	16 / 15
64	3.3916 / 3.3665	3.7971 / 3.7706	8 / 7	63 / 64
256	3.1122 / 3.1085	3.5675 / 3.5644	7 / 10	252 / 252
1024	2.8401 / 2.8376	3.2742 / 3.2706	9 / 10	948 / 949
4096	2.7089 / 2.7050	3.1406 / 3.1403	10 / 11	3320 / 3332

M1/N1: 10 of 10 cells froze. Every lattice size, on both seeds, reached full determinism within a suffix depth of 7–11 characters — k^* 's exact value is not seed-stable, but its finiteness is universal, always well inside the probe's depth limit of 20. The contrast with per-coordinate rounding is not a near-miss; it is total (0/23 there, 10/10 here). Note the freeze is not a collapse into a trivial handful of states: at $K=4096$, over 3,300 distinct codewords remain in active use.

M2/N2: the dial is smooth, 8/8. Logistic val bpc improves strictly at every adjacent step of the ladder on both seeds, with gaps (0.385/0.279/0.272/0.131 at seed 0; 0.403/0.259/0.271/0.133 at seed 1) tens of times the tie band. The disclosed non-monotone worry did not materialize for cost, even though k^* itself is mildly non-monotone in K at seed 0 (the two curves are not the same shape — noted in the journal, not chased).

MQ1/NQ1: the cheap end compiles. $K=4096$ sits 0.0570 bits (seed 0) and 0.0531 bits (seed 1) above the float32 logistic reference — a third of the registered tolerance — while still freezing at $k^*=10/11$. Ridge tells the same story (3.1406/3.1403 against the reference's 3.1073), so the price is not an artifact of the trained instrument. Both halves of the win-win landed at the same K on both seeds; per the pre-stated rule, all three claims promoted unbundled after the second pass.

§5 What it means

Cardinality, not resolution, is what compiles. Per-coordinate rounding makes the state grid coarser but leaves the reachable set effectively unbounded, and the rounding perturbation keeps the crystal from freezing. A fixed codebook of visited states caps the reachable set at K outright — and at every tested K the closed-loop dynamics promptly become a function of a short window of recent input. The result rhymes with the automata-extraction literature, where finite abstractions of trained RNNs are constructed by external algorithms [2]; here the network is the automaton, exactly, by construction, and the question was only what that costs.

What it costs is measured, small, and smooth. The compiled pond at $K=4096$ gives up about a twentieth of a bit of next-character prediction relative to its continuous self, and the price rises predictably — no cliff — as the foothold list shrinks toward 16. For the lab's program, this makes "a real, inspectable finite-state machine" an artifact with a measured price, rather than a hoped-for limit.

Open, and posted honestly. Why does *actually visited* matter, as against an arbitrary codebook of the same size? A random-codebook control is under registered study in the lab's journal and remains unpromoted; under house rules it stays out of print, and this paper makes no claim about it. Also open: how far the near-free regime extends past $K=4096$, and whether the compiled automaton transfers across ponds.

§6 References

1. R. M. Gray, "Vector quantization," *IEEE ASSP Magazine* 1(2), 1984.
 2. G. Weiss, Y. Goldberg, E. Yahav, "Extracting automata from recurrent neural networks using queries and counterexamples," *ICML* 2018.
 3. H. Jaeger, "The 'echo state' approach to analysing and training recurrent neural networks," GMD Report 148, German National Research Center for Information Technology, 2001.
 4. M. Mahoney, "Large text compression benchmark" (text8), mattmahoney.net/dc/textdata.
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§7 Provenance

- **oc1eaba4** — 2026-08-11-caity-goat-3: the foothold-lattice brief; 5 cells, seed 0 (run ledger #16). Registered before running, smoke disclosed (including that both smoke K's froze — flagged as suggestive-only at that scale). The pre-stated EUREKA combination fired; replication queued ahead of the whole queue before any findings entry.
- **14o8ob1b** — 2026-08-11-goat-lattice-rerun1: same 5-K sweep, seed 1 — an independent draw of the base pond, the warmup pond, and the codebook sample (run ledger #17). All three claims hit; promoted unbundled per the pre-stated rules.

All claims judged strictly against the registered wording; 2 seeds $\times$ 5 lattice sizes; replicated before publication.

AuxiLab · findings are pre-registered and replicated before publication

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